

Manar Bouras

Data Science & AI Student

☎ (+213) 0557-334-515 — ✉ manar.bouras@g.enp.edu.dz — in manar-bouras
— 🌐 manar-portfolio — 📍 Algiers, Tamenfouste el marsa



About Me

4th Year Data Science and AI student, passionate about solving real-world problems through **deep learning, computer vision and NLP**. Aspiring to bridge AI research with impactful industry applications.

Education

École Nationale Polytechnique

2024 – Present

M.Eng. in Data Science & Artificial Intelligence – Ranked Top 3

École Nationale Polytechnique

2022 – 2024

Preparatory Classes in Science & Technology – Ranked Top 6

Skills

- **Programming Languages & Databases:** Python, Java, C/C++, Scala, SQL, PostgreSQL
- **Data Science & Artificial Intelligence:** Machine Learning, Deep Learning, NLP, Time-Series Forecasting, Clustering, Predictive Modeling, Reinforcement Learning, Computer Vision
- **Data Engineering & Analytics:** ETL, Data Quality, Metadata Management, Data Governance, Dashboarding
- **Web, Full-Stack Development & Tools:** ReactJS, Django, Flask, REST APIs, Git, Docker, DevOps

Experience

- **Freelance Full-Stack Developer – Various Projects (2023–2025)**
 - **APSEC Startup**
Improved and maintained a full-stack application for risk and accident management.
 - **ENP Rebranding Project (ENP-916 Studio)**
Rebuilt the website of École Nationale Polytechnique for its 100th anniversary using WordPress and PHP.
- **Programs**
 - **ICT Summit Africa HealthTech Convergence Lab**
A program organized by ICT in collaboration with the Ministry of Health (Algeria), focused on bridging health-care and technology to develop digital solutions improving the patient journey.
 - **DSIP 3 (2025) – 2-month program (Djezzy, Huawei, MajestEYE)**
Worked on intelligent segmentation for DjezzyApp users, including data quality pipelines, data governance, metadata management, preprocessing, clustering, and dashboarding.
- **Internships / Jobs in Data Science & AI**
 - **Center of advanced technologies CDTA 2026 – Research internship on Medical Image Registration and Content-Based Image Retrieval (CBIR).** Conducted an extensive literature review on medical image analysis methods and deep learning approaches; contributed to solving a medical image registration challenge using deep learning techniques.
 - **Perrenix 2026 AI & Legacy R&D Engineer** – Designed a secure COBOL-to-Python migration pipeline using **EnterpriseCobolAirlock**, featuring deterministic and reversible PII masking, custom recognizers for COBOL/CICS/DB2, and LLM-assisted translation with full preservation of business logic.
 - **Holcim El-Djazaïr (ex-Lafarge) 2026** – Developed a **Spend Analytics dashboard** and supplier decision-support tool using SAP data (Python, Pandas, Plotly, Streamlit); implemented an end-to-end data pipeline.
 - **Djezzy 2025** – Developed ML models to identify high-potential prepaid clients for postpaid migration.
 - **DSIP 3 2025 – 2-month program (Djezzy, Huawei and MajestEYE)** focused on intelligent segmentation for DjezzyApp users (Data Quality Pipelines, Data Governance, Metadata Management, Preprocessing, Clustering, Dashboarding).
 - **Kabas Startup 2025** – Predictive modeling for drilling fluids using Machine Learning, NLP, PyTorch, and ETL pipelines.
 - **ATM Mobilis 2025** – Built Power BI dashboards and developed a customer churn prediction model.
 - **Heetch DZ 2024** – Gained exposure to industry workflows, team collaboration, and software development practices.
- **Projects**
 - **Retail Intelligence Platform (Big Data & BI)** – Built a retail analytics solution using Hadoop, Spark, SSIS, SSAS, and Power BI, including ETL pipelines, a star-schema data warehouse, OLAP cubes, interactive dashboards, and RFM-based customer segmentation.
 - **Autonomous Vacuum Cleaner Robot** – Developed a ROS 2-based cleaning system on TurtleBot3 with real-time dirt detection (HSV → YOLOv8). Trained on a hybrid dataset (Roboflow + Gazebo) and explored diffusion policy-based imitation learning for adaptive navigation.
 - **Research in Neuro-Symbolic AI for Supply Chain Shortage Management** – Hybrid AI system com-

binning deep time-series forecasting and symbolic optimization to compute robust reorder points under demand variability.

- **NLP Platform for Pharmaceutical Supply Chain** – Built a full NLP pipeline including Speech-to-Text, NER for report generation, Next-word prediction, RAG system, sentiment analysis, text summarization, and spell checker. Deployed using Docker and Hugging Face Spaces (MLOps).
- **Cryptography File Security App** – File Encryption/Decryption using AES, RSA, and DES.
- **Face Recognition Web App** – Developed with Flask and OpenCV.
- **Adaptive Learning System** – Implemented using Random Forests and Reinforcement Learning.
- **Full-Stack AI Website for Hackathon** – Web development, Machine Learning models, and Analytics Dashboard.
- **Lung Nodule Detection** – CNN-based medical imaging project.

Certificates

Data Scientist Career Track (DataCamp, 2025) — Advanced Learning Algorithms (Coursera, 2025)

Extracurriculars

-
- Business Game IEC (2025) – Project Manager & External Relations
 - Vision & Innovation Club (2023–2024) –HR / IT Member / HOST

Awards & Recognitions

2nd Place – DEVCamp Hackathon (2025)

Soft Skills

Teamwork & Collaboration — Communication & Public Speaking — Project & Time Management

Languages

Arabic – Native — **English** – C1 — **French** – C1 — **Korean** – A1